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Artificial intelligence applications in the football codes: A systematic review

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ABSTRACT

Artificial Intelligence (AI) is increasingly being adopted across many domains such as transport, healthcare, defence and sport, with football codes no exception. Though there is a range of potential benefits of AI, concern has also been expressed regarding potential risks. An important first step in ensuring that AI applications in football are usable, beneficial, safe and ethical is to understand the current range of applications, the AI models adopted and their proposed functions. This systematic review aimed to identify different applications of AI across football codes to synthesise current knowledge and determine whether potential risks are being considered. The systematic review included 190 peer-reviewed articles. Nine areas of application were found ranging from athlete evaluation and event detection to match outcome prediction and injury detection and prediction. In total, 27 different AI models were identified, with artificial neural networks the most frequently applied. Five AI assessment metrics were identified including specificity, recall, precision, accuracy and F1-score. Four potential risks were identified, concerning data security, usability, data biases and inappropriate athlete load management. It is concluded that, though a wide range of AI applications currently exist, further work is required to develop AI for football and identify and manage potential risks.

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KEYWORDS
Artificial intelligence; sport; football; risk; safety

Introduction

Artificial Intelligence (AI) is defined as intelligent machines or software which have the ability to mimic human intelligence (Bonaccorso, 2018). AI is rapidly evolving and is becoming increasingly prevalent in our everyday lives. Current AI-based technologies which people interact with daily include Apple's Siri, Google's Alexa, OpenAI's ChatGPT and Tesla's self-driving car (Bubeck et al., 2023; Kaplan & Haenlein, 2019; Naudé & Dimitri, 2020). In association football (soccer), one example of AI technology includes the semi-automated offside technology that was used in the 2022 Qatar World Cup (FIFA, 2023). AI technologies are designed with the aim of making life and work more efficient by automating processes which would otherwise require significant investment of resources and time (Bonaccorso, 2018).

AI has been implemented in a broad set of domains including transport (Crayton & Meier, 2017), healthcare (Sunarti et al., 2021), logistics (Y. Liu et al., 2023), defence (W. David et al., 2020) and education (K. Zhang & Aslan, 2021) to name only a few. If designed and integrated well, AI affords many opportunities and benefits. For example, autonomous vehicles offer safer and more efficient travel (Crayton & Meier, 2017). The implementation of AI for logistics has reduced costs of resources, time and labour by automating tasks allowing human operators to engage in other activities that benefit productivity (Y. Liu et al., 2023). The application of AI as virtual assistant nurses potentially improves the efficiency of healthcare facilities by providing basic diagnosis and supporting effective triage without the need for human nurses or doctors (Crayton & Meier, 2017). AI in education has provided support for labour shortages wherein not enough educators are available to supply the demands of education (K. Zhang & Aslan, 2021). In defence, AI has been used to remove human operators from dangerous roles (W. David et al., 2020). As such, AI has shown promise through its application to solve a broad range of problems and accomplish many different types of tasks across many domains.

As evidenced in other domains, the potential benefits of AI will lead to increased integration into sport. Further, increasing levels of competition and financial reward in sport mean that clubs and associations will seek to adopt AI to gain an advantage. The popularity of sport and specifically football codes such as soccer, American football, Australian rules (AFL) and rugby codes engage significant viewership worldwide and the associated monetary benefits are substantial, for example, the National Football League in America reports revenues of $USD11.98 billion (NBC, 2023), with the English Premier League (EPL) reporting an annual revenue of £5.5 billion (Deloitte, 2023). To gain competitive advantages, organisations have leveraged the potential of data and analytics to gain a competitive edge. For example, an EPL match produces up to 1.4 million data points which include player positioning, player actions and match events (Z. Yang, 2019). Accordingly,

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these datasets are large, unwieldy and difficult for human operators to analyse and interpret manually, especially given the time pressures associated with professional sports (Passfield & Hopker, 2017). As such, leveraging advanced technologies such as AI to increase on-field and off-field performance of players, teams, clubs and organisations is paramount (Herold et al., 2019).

AI has been implemented across a wide range of sport contexts. These applications include, officiating (FIFA, 2022, 2023), injury forecasting (Zone 7, 2023; Van Eetvelde et al., 2021), talent identification (Owen et al., 2022), match analysis (Arbues-Sanguesa et al., 2020; Tuyls et al., 2021), training plans (Skerik et al., 2018) among others. One task AI is performing in sport includes using computer vision to support human analysts by automatically clipping and labelling videos which are used for analysis (Barra et al., 2023). Beyond automation of time- and labour-intensive tasks, AI may provide novel insights into various aspects of sport such as talent identification (Owen et al., 2022), performance indicators (Parmar et al., 2018) and injury prediction (Van Eetvelde et al., 2021). The rise of AI in sport has important implications for the future of sport.

As evidenced by other domains, AI will undoubtedly change the way in which sport systems work, as human operators shift to jobs which involve supervising tasks undertaken by AI. Consequently, in addition to the potential benefits, there are concerns regarding the risks associated with AI (Hancock, 2017, 2022; McLean et al., 2023; Salmon et al., 2024). One concern, for example, is that separating human operators from tasks will result in the decline of skills required to complete these tasks (Bainbridge, 1983; Endsley, 2023). Further risks identified in other domains include biased AI (Sunarti et al., 2021), privacy and security concerns (Shahriar et al., 2023), the removal of human work (Endsley, 2023) and even existential risks (Bostrom & Yudkowsky, 2018; Salmon et al., 2024). An important area of research in the football codes is therefore to identify the potential risks associated with AI so that they can be identified and managed throughout the lifecycle.

Despite previous research on AI in football codes, there has been no synthesis on the range of AI applications being used, the ways in which AI technologies are being evaluated, or on the risks that are emerging during this initial implementation phase of AI into the football codes. Therefore, the aim of this review was to identify and synthesise the peer reviewed literature describing applications of AI in the football codes (soccer, American football, AFL, Rugby codes) to determine how AI is currently being applied and evaluated. Given the reported risks identified with the implementation of AI in other domains, a secondary aim was to determine what is known regarding the potential risks associated with the implementation of AI in football.

Method

Protocol

The systematic review process followed the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines (Moher et al., 2009). PRISMA is an accepted standard that establishes the evidence-based reporting guidelines that ensure the review process is systematic, comprehensive, transparent and replicable. In the current review, the following information was extracted from each article: author(s) and year of publication; the AI model used (e.g., Deep learning, Random forest); the football code studied (e.g., Soccer, American football); the data or participants involved; the function of the AI (e.g., match outcome prediction, athlete evaluation); identified risks related to AI use; the type of evaluation metric used (e.g., F1-score, recall), and outcomes of the AI model were included to identify the applications.

Electronic search

The literature search was performed on five databases (see Table 1) on 19 July 2022 to identify studies which examined AI use in football. To ensure compliance with PRISMA guidelines, citation software was used (Endnote X9, Clarivate

Table 1. Database and initial search results for each database searched.

Database	Search Terms and Limiters	Articles
Scopus	(TITLE-ABS-KEY ("deep learning" OR "Artificial Intelligence" OR "Machine Learning" OR "Neural Network" OR "Nearest Neighbour" OR "Random Forest" OR "Decision Tree" OR "boosting" OR "Learning Algorithm" OR "Bayesian Logistic Regression") AND TITLE-ABS-KEY ("soccer" OR "football" OR "NFL" OR "Rugby" OR "Rugby Union" OR "Rugby League" OR "Rugby 7s" OR "Gridiron" OR "AFL" OR "Aussie Rules"))	943
PubMed	"Deep learning" OR "Artificial Intelligence" OR "Machine Learning" OR "Neural Network" OR "Nearest Neighbour" OR "Random Forest" OR "Decision Tree" OR "boosting" OR "Learning Algorithm" OR "Bayesian Logistic Regression") AND (("soccer" OR "football" OR "NFL" OR "Rugby" OR "Rugby Union" OR "Rugby League" OR "Rugby 7s" OR "Gridiron" OR "AFL" OR "Aussie Rules") AND (English [Filter])	265
Web of Science	("deep learning" OR "Artificial Intelligence" OR "Machine Learning" OR "Neural Network" OR "Nearest Neighbour" OR "Random Forest" OR "Decision Tree" OR "boosting" OR "Learning Algorithm" OR "Bayesian Logistic Regression") (All Fields) AND ("soccer" OR "football" OR "NFL" OR "Rugby" OR "Rugby Union" OR "Rugby League" OR "Rugby 7s" OR "Gridiron" OR "AFL" OR "Aussie Rules") (All Fields)	1174
SportDiscus	"Deep learning" OR "Artificial Intelligence" OR "Machine Learning" OR "Neural Network" OR "Nearest Neighbour" OR "Random Forest" OR "Decision Tree" OR "boosting" OR "Learning Algorithm" OR "Bayesian Logistic Regression" AND "soccer" OR "football" OR "NFL" OR "Rugby" OR "Rugby Union" OR "Rugby League" OR "Rugby 7s" OR "Gridiron" OR "AFL" OR "Aussie Rules"	122
IEEEXplore	("All Metadata": "Deep learning" OR "All Metadata": "Artificial Intelligence" OR "All Metadata": "Machine Learning" OR "All Metadata": "Neural Network" OR "All Metadata": "Nearest Neighbour" OR "All Metadata": "Random Forest" OR "All Metadata"": Decision Tree" OR "All Metadata": "boosting" OR "All Metadata": "Learning Algorithm" OR "All Metadata": "Bayesian Logistic Regression") AND ("All Metadata": "soccer" OR "All Metadata"": football" OR "All Metadata": "NFL" OR "All Metadata": "Rugby" OR "All Metadata": "Rugby Union" OR "All Metadata": "Rugby League" OR "All Metadata": "Rugby 7s" OR "All Metadata": Gridiron" OR "All Metadata": "AFL" OR "All Metadata": "Aussie Rules")	82

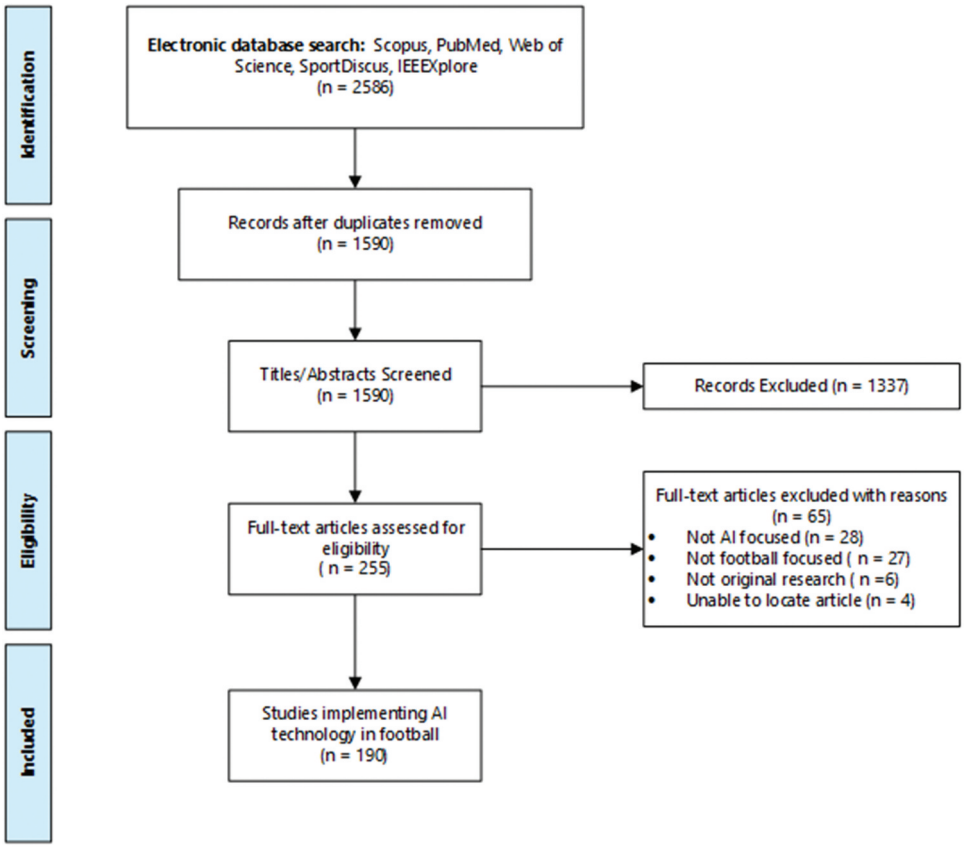

Figure 1. PRISMA flow diagram showing the literature review full-text selection process.

Analytics, Pennsylvania, United States), and at each stage, the inclusion or exclusion of articles was recorded via a reviewing process in Endnote and a purpose-designed spreadsheet. No date limits were applied to the search.

Eligibility criteria

Inclusion criteria

(1) Articles were included if the following criteria was met:
(2) The article described an application or formal evaluation of an AI model in a football code setting (i.e., soccer, rugby codes, Australian rules football, American football)
(3) The article was original research published in a peer-reviewed journal
(4) The article was published in English language

Exclusion criteria

(1) Articles were excluded if any of the following criteria was met:
(2) The article did not include an application of an AI model in a football code setting
(3) The article was a review (Van Eetvelde et al., 2021), white paper (2023), or conference article (Pretorius & Parry, 2016)
(4) The articles discussed perceptions of AI use only (e.g., practitioners' perception of using AI to develop training programs)
(5) The article outlined the development of an AI for football settings (as opposed to an application or formal evaluation of AI in a football setting)

After the search of the databases and removal of duplicates (see Figure 1), titles and abstracts were reviewed against the eligibility criteria, and articles were removed as necessary by the first author. For reliability, a random sample of 10% (159) of the titles and abstracts was uploaded to Covidence a website which assists the conduct systematic literature reviews (Covidence systematic review software, Veritas Health Innovation, Melbourne, Australia. Available at www.covidence.org). The 10% sample of titles and abstracts were reviewed by the second author with 98.11% agreement on included articles All full-text articles were then reviewed and data was extracted by the first author.

Quality appraisal

Quality appraisal is a component of the PRISMA process, however, given that the aim of this systematic review was to identify the broad range of AI applications across football codes to date, appraising the quality of the articles was not required.

Results

Full-text selection

The literature search identified 2,586 articles, with 1,590 remaining after the removal of duplicates. Titles and abstracts

were screened with 1,337 removed and 256 articles remained. During the screening of full-text articles 65 were removed, and 190 articles were included in the final review (Figure 1).

Football codes

The number of articles describing applications of AI in each football code is presented in Figure 2. The included articles covered applications of AI in four football codes, Association football, American football, Rugby codes (union, league and 7's) and Australian Rules Football (AFL). One article examined futsal which was included in the football category. Where an article described applications of AI in multiple football codes, it was counted in each code.

Artificial intelligence applications

Nine different AI application types were identified in the included articles (see Table 2). Applications ranged from player evaluation to injury prediction/detection and event detection to financial applications. The most frequently occurring applications of AI were for player evaluation (44 articles), event detection (40) and match outcome prediction (39).

AI models

In total, 27 different types of AI models were identified. The most frequently occurring types of AI models in the included articles were artificial neural networks (ANN; $n = 35$), decision trees (DT; $n = 23$), extreme gradient boosting (XGBoost; $n = 22$), support vector machines (SVM; $n = 19$), Random Forests (RF; $n = 18$), Deep learning (DL; $n = 18$) and convolutional neural networks (CNN; $n = 17$). An additional 20 AI algorithms were applied with less frequency (see Table 3)

AI evaluation methods

The different evaluation measures that have been used to assess AI applications are presented in Table 4, including both individual measures and various combinations of measures. The following five individual measures were identified in the included articles:

(1) Accuracy. Accuracy represents the total number of correct predictions against all cases in the dataset for categorisation problems. Root mean square error and mean absolute error were also used for analysing predictions made by AI. total
(2) Specificity. Specificity represents the true negative rate and describes how well the AI can detect the negative category available (Bonacorso, 2018).
(3) Sensitivity or recall. Sensitivity represents the true positive rate and describes the AI's ability to detect the positive category available (Bonacorso, 2018).
(4) Precision. Precision represents the quality of true positives identified and is calculated as the number of true

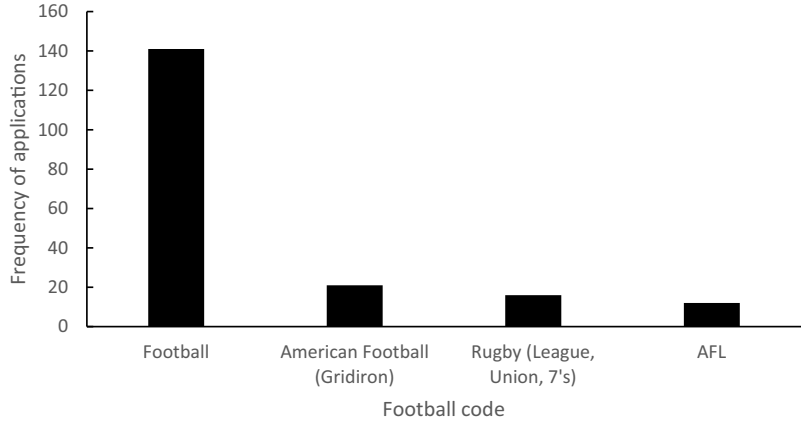

Figure 2. Frequency of football code applications of AI.

Table 2. AI applications, descriptions and frequency in included articles.

Application	Description	*n*
Player evaluation	To determine player ability or playing style, to assess performance indicators, identify talent and assess potential impact in matches	44
Event detection	Player/object tracking and was extended to include highlights such as scoring plays or fouls	40
Match outcome prediction	To determine the outcome of matches, competitions and score lines and to determine key factors associated with successful play and gambling strategies	39
Injury prediction/ detection	To predict injuries (i.e., soft tissue non-contact injuries) or detect and identify injuries (i.e., concussion)	27
Player monitoring – workload	To predict athlete wellness scores related to internal and external load variables	10
Activity prediction	To predict which action will occur in matches (i.e., what type of offence a team will play in American football)	9
Team evaluation	Identified variables associated with successful team performance and identified how teams would play (i.e., line-up, formation)	9
Financial	To predict the value of athletes and sport organisations	8
Match Attendance	To predict the number of patrons who would attend football matches	4

Table 3. AI models applied within football codes.

Algorithm	n
Artificial Neural Network	35
Decision tree	23
Extreme Gradient Boosting	22
Support Vector Machine	19
Random Forests	18
Deep Learning	18
Convolutional Neural Network	17
Ensemble Learning	6
Logistic regression	4
RIPPER (rule-based classification algorithm)	3
Naïve Bayes	3
Self-organising maps	2
Ordinal Regression	2
LASSO	2
k-means clustering	2
Hidden Markov Model	2
Bayesian Network	2
Unknown	1
Ridge Regression	1
Proprietary framework	1
Markov Chain Monte Carlo	1
Linear discriminant analysis	1
K-nearest neighbour	1
Kalman Filter	1
GLMNET	1
Association based classification	1
Additive regression	1

positives divided by the total number of correct and incorrect positive predictions (Bonacorso, 2018).

(5) F1-score. The F1-score is a measure of classification performance and is a combination of precision and recall (Bonacorso, 2018). F1-score provides a metric that handles the limitations of recall and precision by combining their respective strengths.

Accuracy was the most frequently reported metric ($n = 88$). Whilst a single metric provides an easily understandable format to evaluate the performance of AI, many articles applied multiple metrics to provide an in-depth detailed assessment of AI performance (for example: Accuracy * F1-Score * Recall /Sensitivity). Table 4 shows the metrics and combinations of metrics used to assess the performance of AI algorithms, and the number of applications used in each specific combination of metrics in parenthesis.

Identified risks of AI use in football

In the 191 included articles, five reported potential risks associated with the implementation of AI in football. Four categories of risk types were identified (see Table 5).

Discussion

This systematic review synthesised the peer-reviewed literature on applications of AI across the football codes. The findings show that soccer has received the most attention in the peer-reviewed literature compared to other included codes, likely due to its worldwide popularity, and freely available online data (see Kaggle.com; Datahub.io). In total, nine different AI application types were identified, along with 27 different AI algorithms. The following sections provide a summary of the findings in each application area, along with a discussion of the potential opportunities and challenges of AI implementation in football.

Athlete evaluation

The review identified 44 articles which described the use of AI for predicting how athletes will perform during matches, or for identifying talented athletes. AI has been deployed to classify playing ability (Abdullah et al., 2016; Hosp, Schultz, Honer, et al.,

Table 4. Evaluation methods applied in the included articles and applications.

Metric/s used	Total	Application
Accuracy	88	Outcome prediction – match result (25); Player evaluation (20); Event detection (14); Injury prediction/detection (7); Financial (6); Player monitoring – load management (6); Activity prediction (4); Match attendance (3); Team evaluation (3)
Accuracy * F1-Score * Recall/Sensitivity * Precision	21	Event detection (8); Player evaluation (6); Outcome prediction – match result (5); Match Attendance (1); Player monitoring – load management (1)
F1-Score * Recall/Sensitivity * Precision	13	Injury prediction/detection (6); Event detection (4); Outcome prediction – match result (1); Player evaluation (1); Team evaluation (1)
Accuracy * Recall/Sensitivity	8	Event detection (3); Injury prediction/detection (3); Player evaluation (2)
Accuracy * Recall/Sensitivity * Specificity	7	Outcome prediction – match outcome (3); Injury prediction/detection (2); Player evaluation (1); Activity prediction (1)
Accuracy * Recall/Sensitivity * Precision	6	Event detection (3); Outcome prediction – match result (1); Team evaluation (1); Injury prediction/detection (1)
Accuracy * F1-Score	5	Injury prediction/detection (2); Event detection (1); Player evaluation (1); Player monitoring – load management (1)
F1-Score	3	Event detection (1); Outcome prediction – match result (1); Player evaluation (1)
Recall/Sensitivity * Precision	3	Player evaluation (1); Event detection (1); Injury prediction/detection (1)
Accuracy * F1-Score * Recall/Sensitivity * Precision * Specificity	2	Player evaluation (1); Activity prediction (1)
Accuracy * Precision	2	Player evaluation (1); Event detection (1)
Accuracy * Recall/Sensitivity * Precision * Specificity	2	Player evaluation (1); Injury prediction/detection (1)
Accuracy * F1-Score * Recall/Sensitivity	1	Activity prediction (1)
Accuracy * F1-Score * Recall/Sensitivity * Specificity	1	Outcome prediction (match result) (1)
F1-Score * Recall/Sensitivity	1	Event detection (1)
Precision	1	Event detection (1)
Recall/Sensitivity	1	Event prediction (1)
Recall/Sensitivity * Specificity	1	Injury prediction/detection (1)

*Denotes used in combination with other methods.

Table 5. Identified risks of AI use in football.

Risk	Definition	Potential Outcomes
Data security	Issues regarding the security of cloud-based data storage and the potential for access by outside users was considered (Lyubovsky et al., 2022; Wei et al., 2016)	Confidential and sensitive information related to training, tactics, or player information may be leaked to unauthorised parties which may use the data inappropriately or to gain an unfair advantage.
Biased data	AI models learn through the data provided; the inputs have the potential to be biased and produce inappropriate outputs which may cause harm (Bauer & Anzer, 2021)	Outputs may emphasise an inappropriate course of action for example, a specific substitution or tactical change which has a negative impact.
Usability of equipment	AI technology involves complex mathematical algorithms which may be difficult to interpret, further the use of black box modelling limits the interpretability of outputs (Hassan et al., 2020).	Sports practitioners who use AI may be unable to interpret outputs resulting in incorrect decision-making or potentially low uptake of AI for sport.
Inappropriate load management	Implementing AI as a load management and/or injury prevention tool presents potential for athletes to be under or over-used (Vallance et al., 2020).	Incorrectly identified athletes would be at risk of being utilised too frequently resulting in injuries, or potentially underutilised resulting in poor development and maintenance of playing ability.

2021; Hosp, Schultz, Kasneci, et al., 2021; Jamil et al., 2021; Maanijou & Mirroshandel, 2019), assist in team selection (Abidin, 2021; Cortez et al., 2022; Woods et al., 2018), support talent identification (Jauhiainen et al., 2019; Owen et al., 2022) and evaluate players' technical skills (Anzer & Bauer, 2022; Whitehead et al., 2021; Xue et al., 2022). The data used to determine these evaluation outcomes is often collected from physical testing protocols, alongside tracking and monitoring during matches and training (Drust, 2019). Given the broad spectrum of physical and performance metrics collected on athletes and teams in football, organisations may benefit from the increased efficiency afforded by AI.

These data collection and analysis tasks are typically resource intensive, and automation of these processes through AI could enable a more efficient and effective process. However, the review revealed that concerns remain regarding the capability of AI to conduct these tasks effectively. Missing information will limit the applicability of outputs, for example, contextual factors such as field size may influence the passing characteristics and style of athletes (H. Cho et al., 2022). Further, without relevant data for specific activities such as off-ball actions (e.g., defensive actions, positioning, runs into the box), comprehensive, and accurate evaluations cannot be achieved (Li et al., 2022). The implications for practitioners are that athlete evaluations conducted by AI may not consider all relevant and contextual factors and thus must be carefully considered before including the outputs into decision-making processes. Further work exploring the development and evaluation of more comprehensive athlete evaluation AI technologies is therefore recommended. This could be supported by detailed AI models which include the range of functions and attributes required for players in different positions and roles (Berber et al., 2022; McLean et al., 2017).

Event detection

The review identified 40 articles which described AI use to detect discrete actions and events which occur during training and matches. The use of AI to identify events or objects has implications for the way in which analytics is conducted in sport. For example, analysts may use AI to automatically select moments in matches or training, therefore reducing human workload and improving the latency between the action occurring and when the video is available to view. The findings revealed that AI has been applied to identify and categorise a range of football events, for example passing (Chawla et al., 2017; Stoeve et al., 2021), tackles (Chambers et al., 2019), counter pressing (Bauer & Anzer, 2021), goal detection (Elgamml et al., 2020; Muhammad et al., 2021) and catches by American football players (Hollaus et al., 2020). In addition, applications for detecting player actions (i.e., kicking, jumping, running) (Cuperman et al., 2022), ball detection (Ancona et al., 2003), player detection (J. Liu et al., 2009; Maćkowiak, 2013) and target tracking (Zhao & Pan, 2022) were also identified. These events, alongside athlete locomotion such as distance covered, high-speed running and decelerations among many others, form the basis of current analysis techniques, which are considered critical for success (Bauer & Anzer, 2021; Elgamml et al., 2020; Fani et al., 2017).

The review identified that data from training and matches is typically obtained from devices and sensors worn on the body or clothes, GPS and videos (Chambers et al., 2019; Cui, 2018; Rossi et al., 2018). For effective AI output, high-quality input data must be obtained (Buolamwini & Gebru, 2018). Many of the articles indicated that output accuracy was diminished when targets were occluded, sensor errors occurred and low image quality was used for tracking (Jin, 2022). An irony of AI is the increased level of human oversight required to monitor AI outputs to achieve meaningful results, which in turn negates the benefits of automation (Endsley, 2023). Beyond issues with the accuracy of AI outputs, it should be noted that many of the identified AI technologies examined specific actions or events in isolation. Given the importance of interactions between events and actions in football, AI technology and subsequent analysis should consider these elements together to better understand the inherent complexity and emergent behaviours of these interactions within football (Salmon & McLean, 2020).

Outcome prediction

Thirty-nine articles described the use of AI to predict match outcomes. AI model inputs for match prediction included variables such as anthropological measures of athletes, successful passing in the opposition's half, total number of tackles and shots on target, to name a few (Bilek & Ulas, 2019; Bunker & Spencer, 2022; J. A. David et al., 2011). Beyond the application of determining which team would win, AI was also used to elucidate the variables which underlie successful performance across football codes, for example, shots on target (Almulla & Alam, 2020), possession (Bilek & Ulas, 2019), passing and

rushing yards (J. A. David et al., 2011). Further, novel data such as X (formerly Twitter) trends during matches associated with the teams playing (Wunderlich & Memmert, 2022) and the weather (Iskandaryan et al., 2020) was also used to determine outcomes of matches. Higher-order data such as passing networks were also developed to be applied to predict match outcomes (Ievoli et al., 2021).

Whilst outcome prediction is a popular area of research, significant limitations are discussed in the included articles. The accuracy of AI in predicting match outcomes across football codes was low, and the majority of applications did not provide meaningful outputs (Berrar et al., 2019; da Costa et al., 2022). Further, a critical issue is that several studies removed draw results from analyses to improve the accuracy of AI models, creating artificially inflated results (Bennett et al., 2019; Y. Cho et al., 2018; Geurkink et al., 2021; Goes, Brink, et al., 2021; Hassan et al., 2020). Although match prediction is an important area for sport, it is concluded that further work is required to enhance the accuracy of match prediction AI.

Injury prediction/detection

The review identified 27 articles which described AI as an injury prediction or detection tool. Given the negative personnel and organisational impacts of injuries, AI injury prediction and detection tools represent an important area emerging in sport. For example, financial losses associated with injuries with EPL teams on average equate to £45 million per season due to performance decrement and wages paid to injured players (Eliakim et al., 2020). The review revealed that AI has been deployed to predict football code athletes' risk of sustaining injuries such as hamstring strains (Ayala et al., 2019) and concussions (S. Wu et al., 2019). By accurately identifying injury risk, athletes may be able to be managed more efficiently resulting in decreased likelihood of sustaining injuries (Hecksteden et al., 2022). Further automating injury detection is estimated to save up to 12 h of manual video analysis (Raymond et al., 2022).

The findings from this review indicate that injury prediction AI is currently limited. To date, AI has only been developed to categorise athletes as either at high or low risk of sustaining an injury (Carey et al., 2018; Rossi et al., 2018). As such, AI is being used for simplistic forecasting rather than prediction (hence use of the term prediction is misleading). This conceptualisation of injury likelihood is also problematic as it fails to provide a clear estimation or course of action for practitioners (e.g., coaches, physiotherapists, medical staff) who monitor player health and well-being (Hecksteden et al., 2022; Oliver et al., 2020). This issue is compounded further as AI with high false positive rates (low precision) would identify low-risk athletes as high-risk resulting in improper athlete management (Rossi et al., 2018). It is concluded then that the use of AI for injury prediction is limited and is an area that requires further research to develop and establish tools with greater predictive ability.

Assessing how well AI performs is crucial for the implementation of AI in football. However, within the included articles there does not appear to be consensus on the metrics for performance. Transparent and understandable measures of performance are important as they allow stakeholders to make informed decisions regarding the implementation and use of AI (Kumar et al., 2024). If stakeholders are not adequately informed of the boundaries of potential AI, it may result in inappropriate use potentially causing harm to individuals, teams and organisations.

Opportunities for AI in football

The review revealed a broad set of possibilities for applications of AI across football codes. Given the increasing amount of data available in sport, AI provides a solution to manage big data (O'Leary, 2013). Further, elite sport is fast-paced and turnaround times between matches are short, and the ability of AI to perform tasks in real time will reduce the latency between data collection and outputs (Fernández et al., 2021; Wei et al., 2016). This is particularly useful when assessing the health status of athletes, for example, concussions (S. Wu et al., 2019). Further, real-time analysis could improve the decision-making of coaches and performance staff by providing information previously unavailable during matches or training (Fang et al., 2021; Goes, Kempe, et al., 2021).

Beyond the efficiency gains of automation, AI may also create new knowledge through the discovery of previously unidentified patterns in the data. For example, AI was used to expedite the creation of COVID-19 vaccines by identifying antigens faster than human operators (Sharma et al., 2022). In football, AI may be able to identify previously unknown performance metrics to accurately predict performance across football codes (Young et al., 2019), or how best to manage athletes to reduce the occurrence and severity of injuries (Gabbett, 2016). Further, AI capable of integrating diverse data sources, including broader social and environmental factors, may identify patterns or novel metrics to extend current knowledge regarding injury occurrence. Given the exponential rate at which AI is advancing and its proliferation into sport and football, it will likely transform many of the current roles and occupations we currently know (McLean et al., 2022).

Challenges and risks for AI in football

Although a wide range of AI applications and opportunities were identified, there remain risks and challenges to the implementation and use of AI in football. Specifically, the review revealed that four categories of risk have been discussed: issues with data security, data biases, poor usability and inappropriate load management. For example, one risk identified in the review was around biased data (Bauer & Anzer, 2021). As AI learns and begins to make projections based upon the provided data, it is important that the quality and variety of data is considered (Bonaccorso, 2018). A current issue with AI is the "garbage in, garbage out" problem wherein low-quality data lead to poor or biased output results (Buolamwini & Gebru, 2018; Owen et al., 2022). For example, an injury prediction AI trained on low-quality data would likely produce poor outputs which may incorrectly identify athletes' risk or suggest inappropriate changes to training (Vallance et al., 2020); or computer vision which is used for analytics may not identify relevant

clips if match phases or actions are occluded (e.g., single-camera angle) or not clear from the recordings (Jin, 2022). Therefore, future AI should be designed and developed with considerations to appropriate data collection which should include the extraction of new and novel variables (Young et al., 2019).

The usability of AI for stakeholders in sports is a challenge to the implementation and integration of AI in football (Hassan et al., 2020). The development of the models and interfaces which the end-users in football interact with must be carefully designed to improve functioning and limit adverse events (Endsley, 2023). The usability of new technology such as AI is important for practitioners to achieve the proposed benefits (Hammes et al., 2022). For example, end-users may be limited in their ability to determine the limits of the AI and will potentially overvalue the AI outputs during decision-making (Endsley, 2023). By misinterpreting AI outputs there is a risk of an adverse event occurring which may cause harm to the individual, team, or organisation.

Whilst the four categories of risk identified in the literature are encouraging, it should be noted that other additional risks have been found in other domains. These include risks to health and safety (Read et al., 2022), regulatory risks and enfeeblement risks (Salmon et al., 2024) to name a few. It is therefore concluded that further research is required to identify potential risks and identify suitable controls. This work should consider risks across the football AI lifecycle and should use appropriate risk assessment and management methods. It is our view that football researchers and practitioners can learn from other domains where a wide range of AI risks have been identified, including healthcare (Sunarti et al., 2021), transport (Read et al., 2022) and defence (W. David et al., 2020). The notion of AI risk in football is beyond that of just on-field risks and brings about a potential raft of systemic challenges. One example of the implementation of AI in football is employment risks, whereby current roles and jobs in football will be removed, with knock-on effects meaning our current sports science education will likely be redundant and in need of reform. Further, as automation increases, separation of human operators from tasks will result in the decline of skills required to complete these tasks, and the concomitant degradation of job satisfaction (Bainbridge, 1983). Ideally, in the short term, AI will be developed with human-AI teams in mind which work in tandem to optimise tasks thus improving performance without losing human elements (Endsley, 2023, Naughton et al., 2024). As such, a broader understanding of the emergent impacts of AI implementation in football is required.

Future directions

To better understand the risks of AI implementation into sport, future investigation should implement the use of prospective systems thinking-based risk assessment methods to identify the risks associated with AI implementation in football. For example, methods such as Systems-Theoretic Accident Model and Processes Systems-Theoretic Process Analysis (STAMP-STPA; Leveson, 2012), the Networked Hazard Analysis and Risk Management System (Net-HARMS; Dallat et al., 2018) and the Event Analysis of Systemic Teamwork Broken Links approach (EAST-BL; Stanton & Harvey, 2017) have been proposed as useful for identifying systemic AI risks in other safety critical domains (McLean et al., 2023; Salmon et al., 2022, 2023, 2024). Further, the development of a regulatory framework for responsible AI in football that considers issues around data quality, biases and ethics is recommended as a critical and urgent step. Such frameworks and principles have been developed in others such as Defence (e.g., NATO, 2021); however, at present, a unified and accepted regulatory framework for AI implementation in sport has not been forthcoming. Robertson et al. (2023) have developed a framework for technology in sport generally; however, it does not consider AI and its potential impact specifically.

Limitations

The current review aimed to identify the spectrum of AI applications across football codes. However, the development of AI models was considered beyond the scope of this review, as such the findings are limited regarding specific details of the development process. Whilst the algorithms associated with the models are important, the focus of this review was to examine AI applications and the risks identified with the AI lifecycle. A further limitation of this review is that the search was completed in July 2022. Given the rapid proliferation of AI since this time, it should be noted that the current review provides an assessment of applications of AI in football up to that point in time, and does not reflect recent advances such as the introduction of ChatGPT. A future review should be undertaken with a specific focus on the use of Large Language Models (LLMs) in both football and sport.

Conclusion

The use of AI in football is becoming increasingly popular due to the abundance of available data, its many relevant applications and the speed of AI development. The aim of this systematic review was to identify and assess the applications of AI in football codes. A secondary aim was to identify whether risks associated with the implementation of AI in football had been considered. The review identified a broad range of AI applications for football, which present opportunities for the functioning of football systems. However, there are challenges which must be addressed. By taking our knowledge of AI development and implementation in domains outside of sport, AI development and implementation for sport may be enhanced. For example, understanding the important aspects design and risks around its implementation may improve its safety, usability and effectiveness which will enhance its performance and subsequent uptake in practice.

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